A Cartan / Weyl–Chamber Reading of the Composite–Spin Readout

Local–nonlocal–local transport between color holes, and whether the entangling exponential is a gluon

Tessera theory notes (companion to `spin_readout.tex`; part of issue #410)

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Abstract

This is a research–direction companion to `spin_readout.tex` (same directory), which establishes that the proton's composite spin $J^2 = \frac{3}{4}$ is an *entangled*, mixed–symmetry three–body quantity and that any readout reducing each color hole to its own Bloch vector provably floors at the product mixture ($\frac{7}{4}$ per ket, $\sim\frac{9}{4}$ for three). Here we take seriously a single organizing idea: read the relation between two color holes through the *Cartan* (KAK) decomposition $g = k_1\, a\, k_2$, whose structure is exactly *local–nonlocal–local*. Three things fall out. (i) The product floor is precisely the statement that the reconstructed joint state is confined to the K–orbit (the k's), discarding the abelian/entangling middle factor a. (ii) The Weyl group of the color group SU(3) is the symmetric group S_3, and it is the *same* S_3 whose transpositions appear as the swap operators P_{12}, P_{13}, P_{23} in $J^2 = \frac{3}{4} + \sum P_{ij}$; the carried color register $[1, \omega, \omega^2]$ is the Weyl–covariant assignment of weights to holes. (iii) The entangling exponential $\exp(i \sum_a c_a\, \sigma_a \otimes \sigma_a)$ already appears in the codebase — in the abelian (Schwinger/QED) interaction model, where it is explicitly the *neutral photon*. "Acts like a gluon" is then a precise, falsifiable proposal: occupy the same mediator slot with the *non–abelian*, color–valued, geometry–sourced version. We state what would make that true, what is merely analogy, and a concrete next experiment. Nothing here is claimed as established; it is a lens and a program.

Contents

1 What this companion adds, in one paragraph

`spin_readout.tex` ends at the *total–space spin operator*: to recover $\frac{3}{4}$ the spin Casimir must act on the whole carried representative at once, keeping the cross–hole terms. That is correct, and we do not retract it. What this note adds is a *name* for the missing object and a *coordinate* for it. The name is the abelian middle factor a of a Cartan decomposition; the coordinate is its Weyl–chamber position (equivalently, the local–unitary invariants of the two–hole state). Naming it buys three concrete things: a sharper diagnosis of why the product read floors (§2), an identity — not an analogy — between the color group's Weyl symmetry and the swap operators that produce $\frac{3}{4}$ (§3), and a precise version of the user's question "does the exponential act like a gluon?" (§4). Two payoffs (§5) and the honest caveats (§6) follow.

2 The Cartan (KAK) decomposition and the entangling core

The decomposition. Let G be a compact Lie group with Cartan involution θ, splitting the algebra as $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ ($[\mathfrak{k},\mathfrak{k}] \subseteq \mathfrak{k}$, $[\mathfrak{k},\mathfrak{p}] \subseteq \mathfrak{p}$, $[\mathfrak{p},\mathfrak{p}] \subseteq \mathfrak{k}$). Choose a maximal abelian $\mathfrak{a} \subseteq \mathfrak{p}$. The *Cartan decomposition* (the KAK decomposition) writes every $g \in G$ as

$$g = k_1\, a\, k_2, \qquad k_1, k_2 \in K = \exp \mathfrak{k}, \quad a \in A = \exp \mathfrak{a}. \tag{1}$$

The A factor is determined up to the *Weyl group* $\mathcal{W} = N_K(\mathfrak{a})/Z_K(\mathfrak{a})$, and a fundamental domain for $\mathcal{W}$ acting on $\mathfrak{a}$ is a *Weyl chamber*. The chamber coordinates of a are the only data of g invariant under $g \mapsto k_1' g\, k_2'$ — the genuinely two–sided, frame–independent content.

The two–qubit instance (the one we use). For two spin–$\frac{1}{2}$ holes, $G = \mathrm{SU}(4)$, $K = \mathrm{SU}(2) \times \mathrm{SU}(2)$, and (1) is the standard "canonical" / KAK form of a two–qubit gate [1, 2]:

$$\boxed{U = \underbrace{(k_1 \otimes k_2)}_{\text{local}}\ \underbrace{\exp\Big(i \textstyle\sum_{a=1}^{3} c_a\, \sigma_a \otimes \sigma_a\Big)}_{\text{nonlocal} = A}\ \underbrace{(k_3 \otimes k_4)}_{\text{local}}} \tag{2}$$

with $k_i \in \mathrm{SU}(2)$ acting on a single hole and (c_1, c_2, c_3) the Weyl–chamber point ($\frac{\pi}{4} \geq c_1 \geq c_2 \geq |c_3|$). The structure is exactly the *local–nonlocal–local* pattern of the user's intuition. The load–bearing theorem:

Claim 1 (Only A can change entanglement)**.** *Local unitaries $k \otimes k'$ preserve the entanglement (Schmidt spectrum) of any two–hole state. Hence the entanglement created by U depends on U only through its chamber point (c_1, c_2, c_3), i.e. through A alone.*

The product floor, in chamber language. `spin_readout.tex` (and `joint_proton_spin_findings.md`, #489) establish that the J^2 *operator* is correct — fed clean states by hand it returns $\frac{3}{4}$, $\frac{7}{4}$, $\frac{15}{4}$ (regression guard in `test_dk_joint_spin.py`). The floor is therefore on the *state* side: `composite_j2` reconstructs the joint spin state as

$$|\Psi\rangle_{\text{read}} = |s_1\rangle \otimes |s_2\rangle \otimes |s_3\rangle\,, \qquad |s_h\rangle = \texttt{emergent\_spinor}(h), \tag{3}$$

a literal `np.kron` of three *independently* extracted per–hole spinors. By Claim 1 a product is fixed at the trivial chamber corner $c = (0, 0, 0)$ (the K–orbit of a product), and no local post–processing moves it off; the entangled $J = \frac{1}{2}$ eigenstate lives in a different orbit. So $\langle J^2 \rangle_{\text{read}}$ is pinned in the three–spin–$\frac{1}{2}$ *product* range $[\frac{3}{2}, \frac{15}{4}]$, clustering near $\frac{9}{4}$ — exactly the observed floor. **The missing ingredient is the A factor in the state reconstruction**: the inter–hole correlations the build put in and the per–hole extraction takes out. (This is the same conclusion as the total–space operator of `spin_readout.tex`; the chamber language just makes the discarded object a named coordinate.)

Remark 1. *The single swap P_{ij} is itself* maximally nonlocal *— it sits at the far chamber corner $c = (\frac{\pi}{4}, \frac{\pi}{4}, \frac{\pi}{4})$ (three CNOTs to synthesize). Note carefully: a swap is maximally nonlocal yet has zero* entangling power *(it maps products to products). This is consistent with the fact that $\langle J^2\rangle$ on the product $|uud\rangle$ is the definite number $\frac{7}{4}$: evaluating J^2 on a product needs no entanglement; reaching the $\frac{3}{4}$* eigenstate *does. The chamber distinguishes the two notions cleanly, which is one reason the language is worth importing.*

Remark 2 (Two distinct "Weyl"s; the chiral one carries the complex deficit)**.** *"Weyl" is used in two unrelated senses, worth separating before §3 leans on the first. The Weyl* group/chamber *is the S_n of the entangling coordinate above (and the S_3 of the color register). Distinct from it is the Weyl* chirality *split of the* single–fiber *frame rotation: in the Euclidean per–cell embedding* $\mathrm{Spin}(4) = \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$, *and with $P_\pm = \frac{1}{2}(\mathbb{1} \pm \gamma_5)$ the structural spin generators are block–diagonal in chirality. The* physical *spatial spin is the* diagonal SU(2) *— the same $\frac{1}{2}\sigma_a$ on both blocks — so writing a connector's exponent as $\vec{c} = \vec{\theta} \pm i\vec{\beta}$, the real part $\vec{\theta}$ is the rotation (spin) and the imaginary part $\vec{\beta}$ is the Lorentzian deficit/boost (the* Im *of $\vec{c}$ is the same physics as the* Im *of the complex Regge action). This is entirely on the K–type spin–fiber side: by Claim 1 it does* not *entangle. Its role is definitional — how the per–hole spin generator is fixed consistently with the complex action — leaving the entangling content where §3–4 put it: the color A factor, not this chiral split.*

3 The Weyl group of $\mathrm{SU}(3)_{\text{color}}$ *is* the S_3 of the J^2 swaps

This section is the one place the Cartan picture is more than analogy. The claim is an equality of groups, with one physical input (Pauli antisymmetry).

Three standard facts.

1. The Weyl group of $\mathrm{SU}(n)$ (root system A_{n-1}) is the symmetric group S_n. For color, $\mathcal{W}(\mathrm{SU}(3)) = S_3$, order 6.
2. The fundamental rep **3** has three weights μ_1, μ_2, μ_3 at the corners of a triangle in the weight plane, with $\sum_i \mu_i = 0$; $\mathcal{W} = S_3$ permutes them. The three color holes carry one weight each.
3. The swap form of the spin Casimir (Eq. (j2swap) of `spin_readout.tex`),
$$J^2 = \tfrac{3}{4}\mathbb{1} + \big(P_{12} + P_{13} + P_{23}\big), \tag{4}$$
is built from the three transpositions of S_3 — its Weyl reflections.

The carried register is the Weyl–covariant assignment. The codebase already realizes fact (2): the carried representative $[1, \omega, \omega^2]$ ($\omega = e^{2\pi i/3}$, `EigenstateSynthesis(st,3).carriedRepresentative`) is *not* $[1, \omega, \omega^2]$ placed by hand but the ω–eigenvector of the window–cycling generator g, the $\mathbb{Z}_3 \subset S_3$ rotation of the color weights (`proton_symmetric_windows.tex`, #398):
$$P_{\text{in}}\,\phi = \omega\,\phi, \qquad M\,P_{\text{in}} = P_{\text{out}}\,M, \tag{5}$$

where M is the inter–hole transport and $P_{\rm in}, P_{\rm out}$ are the (signed) color 3–cycles on the input/output hole triples. Equation (5) says M *intertwines* the color $\mathbb{Z}_3$ — exactly the statement that M carries the Weyl rotation between holes.

The locking (the one physical input). On the *physical* baryon the position–permutation S_3 and the color–Weyl S_3 are not two groups but one. Pauli antisymmetry forces color into the antisymmetric singlet ϵ_{abc} (Eq. (colorsinglet) of `spin_readout.tex`); antisymmetry under exchanging the *positions* of two quarks therefore acts on their *colors* as the corresponding weight transposition. Hence:

$$\boxed{\text{the swaps } P_{ij} \text{ in (4) are the Weyl reflections of } \mathrm{SU}(3)_{\rm color} \text{ acting on the carried register.}} \qquad (6)$$

The $2-1-1=0$ cancellation that yields $\frac{3}{4}$ (`spin_readout.tex`, §3) is the S_3–Weyl structure of the color register read in the spin sector. This is why "a Weyl/Cartan move from one hole to the next" is the right language: the off–diagonal root generators $E_{\pm\alpha}$ of SU(3) — the color–raising/lowering directions, the genuinely non–abelian gluon directions — are precisely what cycle a hole's color $r \to g \to b$, and M in (5) is the existing discrete seed of that action.

4 Does the exponential act like a gluon?

Short answer: it occupies the right *slot*, but in the codebase that slot is currently filled by the *photon*, and the upgrade to a gluon is a specific, non–trivial list.

The photon is already there. The KAK decomposition (2) is implemented in the quantum/interaction subsystem (`src/simulations/InteractionSimulation.cpp`). The comment at the construction is explicit:

> *Cartan/local–frame model: under the KAK decomposition* $U = (K_1 \otimes K_2) \cdot \exp(i\, c\, \sigma\sigma) \cdot (K_3 \otimes K_4)$,
> ... $\Sigma_{AB} \equiv \exp(i\, c\, \sigma\sigma)$ *is the entangling core.*

and that entangling core is identified physically as "*the entangling–core analog of a neutral photon*" with charge $q_{AB} = 0$. So the local–nonlocal–local exponential is *already* playing the role of a force mediator in the engine — as the abelian, neutral one. The user's question is whether the same slot, in the color sector, is a gluon. That is the right generalization to ask, and it is sharp because the abelian case is concrete.

What would make it a gluon (and not a photon). Three conditions, each a real gap:

1. **Color–valued.** The current $\sigma\sigma$ acts on the spin/charge qubits. A gluon's exponential must act on the *color register* — the b_3 harmonic sector carrying $[1, \omega, \omega^2]$ — a different bundle from the spin fiber.
2. **Non–abelian (root, not Cartan–diagonal).** A photon is the abelian direction; a gluon is built from the off–diagonal roots $E_{\pm\alpha}$, so its holonomy is *path–ordered* and non–commuting and reproduces the $\mathbb{Z}_3/S_3$ color cycling around a loop (cf. M in (5)). The neutral $q_{AB} = 0$ core is, by contrast, abelian.

3. **Geometry–sourced, on the right connection.** The existing inter–hole Wilson line (`wilson_line`, `facet_transport`) is the Spin(4) *spin connection* — the deficit–angle, gravity–like frame alignment (`dk_spin_wilson.py`). It is K–type (local, non–entangling) and lives on the *frame* bundle, not the color one. A gluon needs a *color* connection on the register, whose holonomy supplies the chamber point c. **Do not conflate the two**: aligning spinor frames can never entangle (Claim 1); that is the floor, not the cure.

Bare gluon vs. effective hyperfine — the honest interpretation. Even granted (1)–(3), $\exp(i\sum_a c_a\,\sigma_a\otimes\sigma_a)$ most closely resembles not the fundamental colored gluon field (a color–octet vector) but the *one–gluon–exchange color–magnetic hyperfine* interaction [4]

$$H_{\mathrm{hyp}} \;\propto\; \sum_{i<j} (\boldsymbol{\lambda}_i\cdot\boldsymbol{\lambda}_j)\,(\boldsymbol{S}_i\cdot\boldsymbol{S}_j)\,\big/(m_i m_j), \tag{7}$$

with $\boldsymbol{\lambda}$ the Gell–Mann color generators. On a color singlet $\langle\boldsymbol{\lambda}_i\cdot\boldsymbol{\lambda}_j\rangle$ is a fixed constant, so the spin dependence is $\propto\sum_{i<j}\boldsymbol{S}_i\cdot\boldsymbol{S}_j$, which is exactly the operator that splits the nucleon from the Δ:

$$\textstyle\sum_{i<j}\boldsymbol{S}_i\cdot\boldsymbol{S}_j = \frac{1}{2}\big(J^2-\frac{9}{4}\big) = \begin{cases} -\frac{3}{4} & \text{proton } (J^2=\frac{3}{4}),\\ +\frac{3}{4} & \Delta\ (J^2=\frac{15}{4}).\end{cases} \tag{8}$$

The $\sigma_a\otimes\sigma_a$ generator of the chamber *is* the spin–spin coupling of (7). So the precise, defensible statement is: the entangling exponential acts like the *gluon–induced spin–spin (hyperfine) interaction* — the very term whose sign in (8) *is* the proton/Δ distinction. In an *emergent* theory this is the right target: an effective gluon, sourced by register holonomy, would be a *derivation* of the hyperfine interaction from geometry rather than a postulate. Calling it "a gluon" is justified up to that "effective/emergent" qualifier; calling it the bare colored field is not.

5 Two payoffs that are actually usable

(a) The chamber coordinate is the gauge–invariant observable we want. `composite_proton_spin_findi` (#485) makes GAUGE– and RELABEL–invariance the mandatory gates for any J^2. The Weyl–chamber point (c_1,c_2,c_3) — equivalently the Makhlin local invariants [3] — is *by construction* invariant under per–hole frame changes $U\mapsto(k\otimes k')U(k''\otimes k''')$. So it is automatically the kind of observable those gates demand. Concretely, the chamber content of a hole pair is carried by the *connected* two–hole correlator

$$C_{ij} := \langle\boldsymbol{S}_i\cdot\boldsymbol{S}_j\rangle - \langle\boldsymbol{S}_i\rangle\cdot\langle\boldsymbol{S}_j\rangle, \tag{9}$$

which vanishes identically on a product (the floor: $C_{ij}=0$) and is precisely what the per–hole extraction throws away. Since

$$J^2 = \tfrac{9}{4} + 2\sum_{i<j}\langle\boldsymbol{S}_i\cdot\boldsymbol{S}_j\rangle = \tfrac{9}{4} + 2\sum_{i<j}\big(\langle\boldsymbol{S}_i\rangle\cdot\langle\boldsymbol{S}_j\rangle + C_{ij}\big), \tag{10}$$

restoring the C_{ij} restores $\frac{3}{4}$. This suggests a readout that may be *lighter* than the full total–space operator of `spin_readout.tex`: read the three *pairwise joint reduced states* ρ_{ij} off the single carried field and extract C_{ij}, rather than assembling one 16×16 operator on the joint field. The quantum subsystem already stores two–body joints (`jointAB`/`rhoAB` in `InteractionSimulation.cpp`); the machinery exists, and only the proton readout currently replaces it with a `kron`. *Caveat:* this is a sibling of the total–space plan, not a free pass — it still requires ρ_{ij} to be the faithful joint two–hole state, which is the same hard kernel (the Whitney / Kähler–Atiyah fiber$\leftrightarrow$cells lift).

Empirical support already on file. `composite_proton_spin_findings.md` reports that the joint–state prototype (`joint_*`: per–hole spins read from the *single* `carriedRepresentative`$([h_0, h_1, h_2], [1, \omega, \omega^2])$ rather than three independent extractions) is genuinely different from the product, "leans toward the proton channel" (J^2 drops $1.80 \to 1.64$ on the cached structure as the 120° color phases impose $\sim$120° relative spins), and can produce a *definite* Δ ($J^2 = 3.75$) where the product cannot. That is exactly the signature of beginning to keep the A factor: retaining the joint object moves J^2 off the product corner toward $\frac{3}{4}$. (It is not yet a clean observable — it inherits relabel–sensitivity from multi–hole `carriedRepresentative` — which is the open work, not a refutation.)

(b) Promote the transport intertwiner to the A factor. M in (5) already implements the color $\mathbb{Z}_3$ rotation between holes; today it is used only as a *diagnostic* (the intertwining residual $\sim 4 \times 10^{-2}$). The proposal is to use M (or the full color holonomy it generates) as the *source* of the entangling exponential in the state reconstruction — the discrete realization of the off–diagonal–root transport of §3. This is the concrete bridge from "we have a color–cycling transport" to "we have a gluon–like A factor."

6 Honest caveats and where the lens does *not* help

1. **KAK is two–party; the proton is three.** There is no single three–party Cartan decomposition with one Weyl chamber; three–qubit entanglement is genuinely richer (the GHZ vs. W SLOCC classes). The clean realization is therefore a *pairwise / diquark sequence* — entangle the like (uu) pair into the spin–triplet, then couple the d — which is exactly the two–step build already on record ($q+q \to$ diquark, diquark$+q \to$ proton, #489). Good: it matches the build. Bad: "one Cartan decomposition of the proton" is not a literal object; the payoff in §5(a) is pairwise for this reason.

2. **Do not set c by hand.** Choosing the chamber point that forces $\frac{3}{4}$ would be inserting the answer by fiat — the same mistake that had `relax_singlet` removed for placing $[1, \omega, \omega^2]$ by hand. The discipline: the c_a must be *sourced* from the emergent color holonomy (payoff (b)) and the relaxed geometry then *checked* against the proton vs. Δ chamber point. The lens supplies a diagnosis and the right invariant; it does not license hand–tuning.

3. **It does not remove the hard assembly — a different axis.** Two orthogonal problems are easy to conflate, so name them. The fiber$\leftrightarrow$cells lift is *vertical*: at one location, gather ψ's degree–stratified components (the degree–$0, 1, 2, 3, 4$ pieces sitting on different simplices) into the single 16–dimensional DK fiber where Σ_a is even defined — the Whitney / Kähler–Atiyah interpolation, a *local* operation building the fiber over a point. The Cartan / nonlocal content is *horizontal*: across *different* holes, combine those fibers while keeping the cross–correlations — the A factor. KAK is a purely horizontal tool; fiber$\leftrightarrow$cells is the vertical one, and solving the horizontal does not touch the vertical. Worse for any hope that it might: in the operator route $J^2 = \sum_a (\sum_h P_h \Sigma_a P_h)^2$ the vertical is logically *prior* — one cannot even write $\Sigma_a P_h$ (the spin generator restricted to a hole) until Σ_a is defined on the assembled field, so the entangling cross terms sit *downstream* of the lift. In code, `emergent_spinor` already performs a degree–3–only slice of the vertical lift (least–squares the tetrahedral trivector minors into the Clifford algebra), which is why $|\langle \boldsymbol{S} \rangle| = \frac{1}{2}$ per hole is robust; the gap is the *all–degree, jointly–across–holes* version. Reading two holes independently gives $\rho_i \otimes \rho_j$, so $C_{ij} = 0$ by construction. The pairwise–C_{ij} route therefore *reduces* the burden (a two–hole joint lift instead of the full three–hole all–degree global lift) but does not remove it — unless the escape hatch below holds.

4. **Spin connection $\neq$ color connection.** The only inter–hole connection currently in the mesh is the Spin(4) spin connection (gravity–like, K–type). The gluon program presupposes a *color* connection on the b_3 register that does not yet exist as a first–class object; (5)'s M is the nearest seed.

5. **A vielbein alone is the floor.** The user's phrase "vielbein from one hole to the next" should be read carefully: a vielbein is a local frame (K, the `cell_frame`); a pure frame change can never entangle. The new physics is the A factor *sandwiched between* the vielbeine, not the vielbeine themselves. Thinking "vielbein" alone rebuilds exactly the product floor.

The one escape hatch. There is exactly one way the horizontal (nonlocal) perspective could *sidestep* — not solve — the vertical assembly: read the spin content from a *holonomy* rather than from a reconstructed point–fiber. A Wilson loop or the intertwiner M of (5) is intrinsically a *path* object, computed from transport, never from gathering a 16–component column at a point (this is the `dk_spin_wilson.py` / #477 thread, "a frame–independent, register–specific spin holonomy"). The hatch is open **iff** the connected correlator C_{ij} is a *holonomy invariant* — expressible purely from the chamber coordinates of the inter–hole transport (the color $\mathbb{Z}_3$ phases of M, the spin eigenphases of the loop), with no contraction against endpoint spinors. If it is, J^2 can be read off transport data and the point–fiber assembly is avoided; if it is not, a (reduced, two–hole) vertical lift remains. Which of these holds is a concrete, decidable question, set as the core test of §7.

7 A concrete, falsifiable next experiment

In increasing order of commitment:

1. **Pairwise–C_{ij} readout against the validated instrument.** Implement (9)–style pairwise correlators on the joint reduced states and confirm, against the hand–fed clean states already in `test_dk_joint_spin.py`, that they return $\frac{3}{4}$, $\frac{7}{4}$, $\frac{15}{4}$ for proton / product / Δ. This is cheap and either reproduces the instrument in chamber coordinates or falsifies the framing immediately.

2. **C_{ij} two ways — the escape–hatch test.** On one fixture, compute C_{ij} *(a)* from the reconstructed joint two–hole reduced state ρ_{ij} (the vertical route) and *(b)* from the M/Wilson–loop holonomy invariants alone (the horizontal route), and compare. Agreement certifies that C_{ij} is a holonomy invariant — the escape hatch is open and the point–fiber assembly is avoidable; disagreement quantifies exactly how much vertical lift the holonomy cannot replace. Either outcome is a decisive, reportable result.

3. **Read C_{ij} off the emergent joint field.** Apply the same readout to the `joint_*` carried representative and check whether $C_{ij} \neq 0$ and whether J^2 moves further toward $\frac{3}{4}$ than the $1.80 \to 1.64$ already seen — with the GAUGE/RELABEL gates enforced.

4. **Source c from the color holonomy.** Define the color connection whose discrete transport is M of (5), measure its loop holonomy, map it to a chamber point, and test whether the relaxed proton geometry lands on the proton corner and the Δ geometry on the Δ corner — *without* the chamber point being put in by hand. Success here is the operational meaning of "the exponential acts like a gluon."

One sentence. The Cartan/Weyl lens does not by itself produce $\frac{3}{4}$, but it names the missing object (the abelian factor A), gives it a gauge–invariant coordinate (the Weyl chamber / C_{ij}), identifies the color group's Weyl S_3 with the very swaps that produce $\frac{3}{4}$, and turns "is it a gluon?" into a testable program: source that coordinate from an emergent color holonomy and check it lands on the proton corner.

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